

**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2008
Higher 2**

CANDIDATE NAME

CLASS

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INDEX NUMBER

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BIOLOGY

9747/02

Paper 2 Core Paper

9 September 2008

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX no. and name on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions

Section B

Answer any **one** question.

Write your answers on separate answer paper provided.

At the end of the examinations,

1. hand in section A and the 2 questions you attempted from section B separately;
2. fasten all your work securely;
3. enter the number of the section B questions you have answered in the grid opposite.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

Section A

Answer all questions in this section

- 1 **Figure 1** below shows a transmembrane protein that is involved in ATP synthesis.

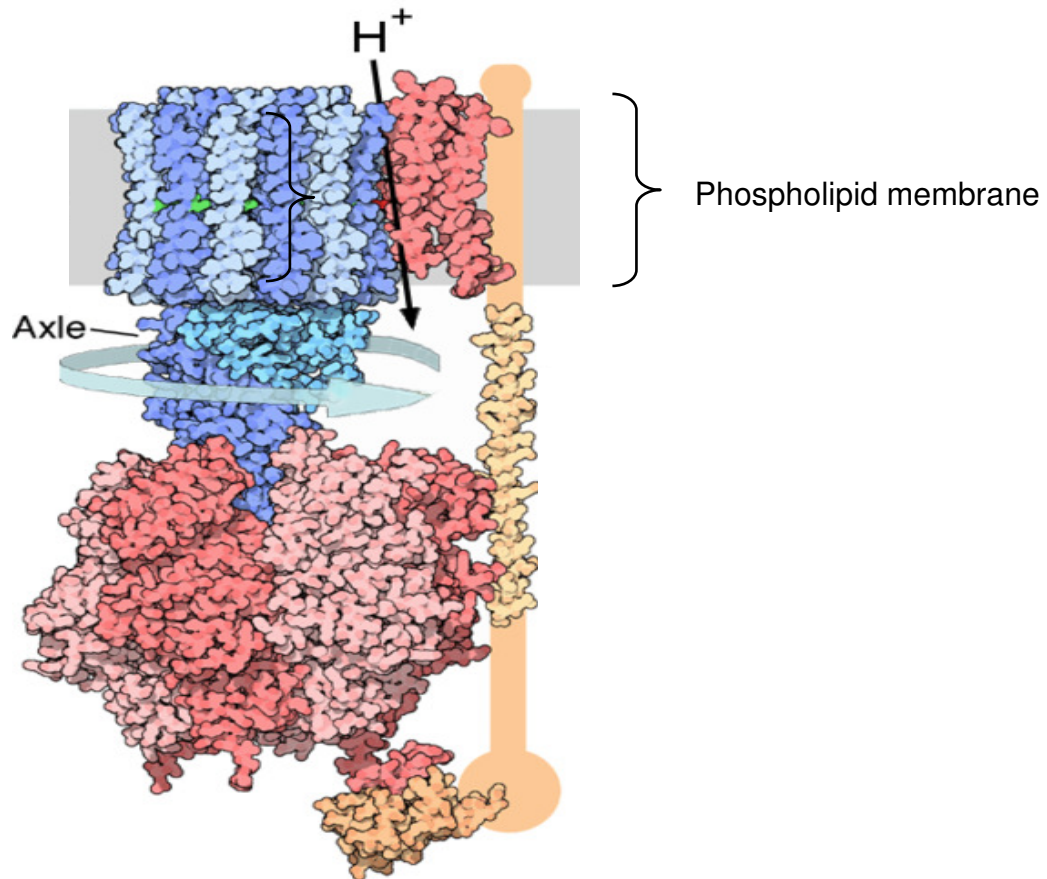


Figure 1

- (a) (i) State **two** specific locations in the cell organelles where this protein may be found.

1.
2.

[1]

- (ii) With reference to the information provided, explain how the specific structure of the protein is determined.

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.....[4]

- (b) (i) With reference to **Figure 1**, explain how ATP is synthesized.

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.....[3]

- (ii) Explain why increasing ADP concentration will increase the rate of ATP synthesis.

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.....[2]

- (iii) (ii) With reference to the concept of limiting factor, explain the trend of the rate of ATP synthesis with further increase in ADP concentration to a high concentration.

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.....[2]

- (b) Complete the table below to show the major differences between the structure of DNA and a fibrous protein. [3]

DNA	Fibrous Protein
1)	
.....	
.....	
2)	
.....	
.....	
3)	
.....	
.....	

[Total: 15]

- 2** One of the earliest drugs used to treat patients with HIV infections was the nucleotide analog AZT, which has a structure similar to a nucleotide. The AZT nucleotide analog being used to treat HIV infections is a dideoxy nucleotide. Dideoxy nucleotides include a ribose sugar that lacks both a 2' and 3' hydroxyl group.

- (a)** Based on what you know about nucleic acid synthesis, explain the effect these analogs have on viral nucleic acid synthesis.

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.....[2]

- (b)** Viruses make use of host cell machinery to synthesise viral protein. Describe the how the structure of a named non-membrane bound organelle enables the organelle to carry out its function of protein synthesis within a cell.

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.....[3]

- (c)** Some individuals are resistant to HIV infection due to an inherited condition where there is a frame-shift mutation in a certain gene that codes for a membrane protein that facilitates entry of the HIV virus into the host cells.

Explain what is meant by a frame-shift mutation?

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.....
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.....[2]

- (d) In some diseases, the changes in DNA sequences often result in dramatic effects in individuals. Using the example of sickle cell anaemia, explain how genotype is linked to phenotype.

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.....[3]

[Total: 10]

3. With the completion of the Human Genome Project, it was estimated that the human genome was likely to contain only 30,000 genes. Counting the number of protein-encoding genes in the human genome is a difficult process, as the “start” and “stop” boundaries of transcribed genes are only imperfectly understood.

A young scientist has decided to take a different approach of capturing mRNAs from the cytoplasm, before converting them to DNA to estimate the number of genes present. Using this approach, he reports that there are over 65,000 discrete mRNA-transcribing sequences in the genome, hence there are more genes than estimated by the Human Genome Project.

- (a) Describe briefly how post-transcriptional modification would allow all types of mRNA to be captured using a single method, and how mRNA can be subsequently converted to DNA.

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.....[2]

- (b) State why counting the number of unique mRNA sequences will result in an over-estimate of the number of actual genes in an organism.

.....
[1]

- (c) Within the same human cell, the level of gene expression can differ greatly between genes. Briefly describe what could account for the different levels of mRNA produced for different genes.

.....

[2]

- (d) The eukaryote genome differs from the prokaryote genome in that some genes may be present in multiple copies, a result of gene amplification.

- (i) Describe briefly one process by which gene amplification can occur.

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[2]

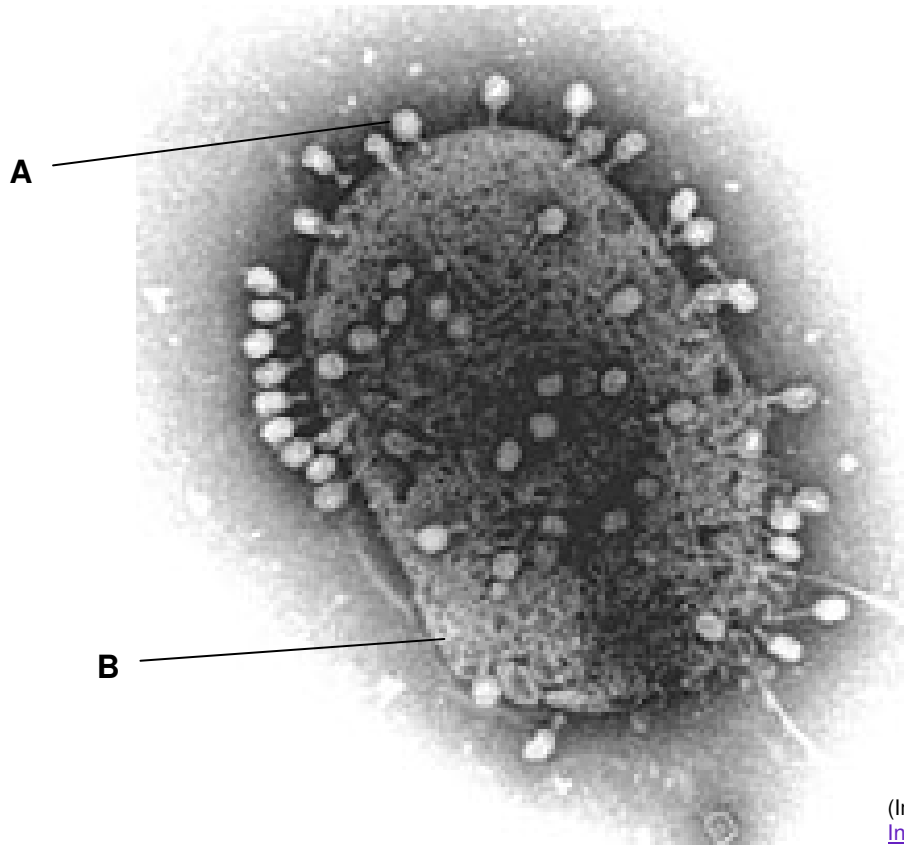
- (ii) In nature, some organisms are able to amplify certain genes at some stage in their development. Describe the significance of gene amplification in development.

.....

[2]

[Total: 9]

- 4 **Figure 4.1** shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.



(Image courtesy of [Cornell Integrated Microscopy Center](#))

Figure 4.1

- (a) (i) Suggest how **A** recognizes the correct host cell.

.....
[1]

- (ii) With reference to **Figure 4.1**, describe a process whereby bacterial DNA can be transferred from one cell to another.

.....

[3]

(iii) In the table below, give two differences between **2 other processes** (besides that one named in (a)) that result in genetic variation in bacteria. [3]

Named Process 1: 	Named Process 2 :
1)	
2)	

Figure 4.2 shows the Jacob and Monod model of gene expression in *Escherichia coli*, the lac operon model.

(Source: <http://www.ccrnp.ncifcrf.gov/~toms/genetic.control/www.uic.edu/lect15.htm>)

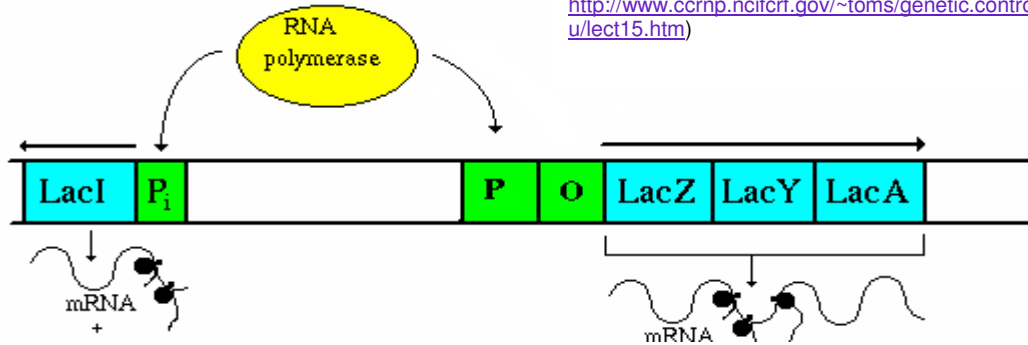


Figure 4.2

(b) Explain the following terms:

(i) RNA polymerase.

.....
[1]

(ii) Operator O.

.....
[1]

(c) With reference to **Figure 4.2**, distinguish clearly between a regulatory gene and a structural gene.

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[2]

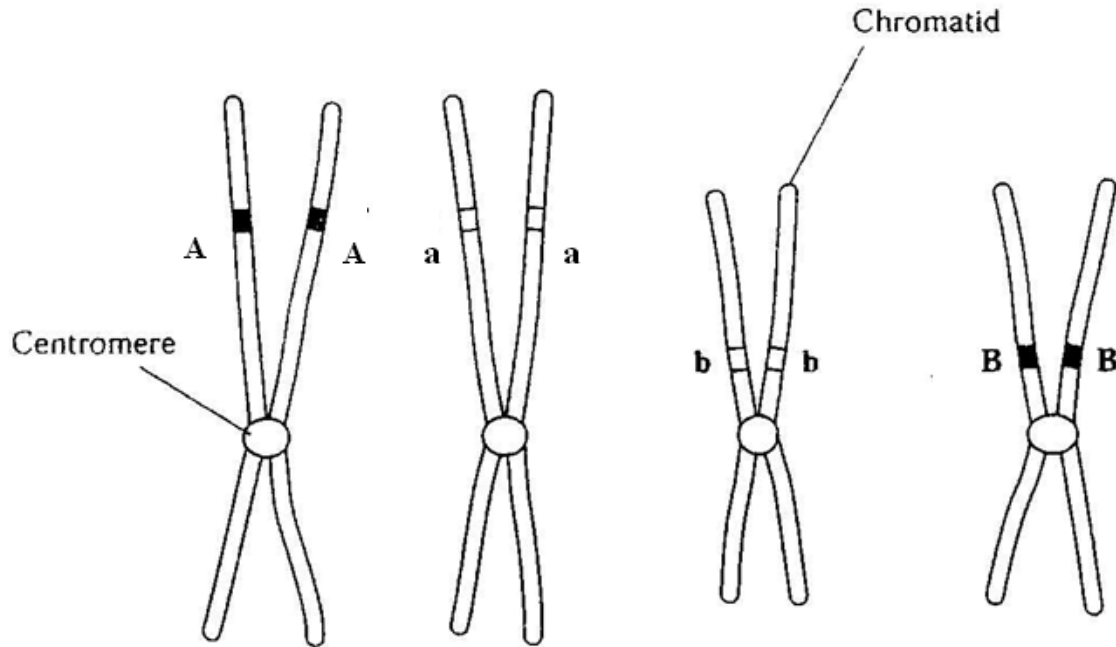
(d) Are the enzymes produced by the lac operon inducible or repressible? Explain your answer.

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[2]

[Total: 13]

- 5 The diagram shows the chromosome from a plant cell in which $2n = 4$. The dominant allele **A** of a gene results in red flower and pink spines on the fruit. The recessive allele, **a**, gives yellow flower and green spines. The dominant allele **B** intensifies the colour of the pink spines to red. The plant is heterozygous as shown below:



- (a) In the space below, draw a diagram to show the appearance of these chromosomes just as they are beginning to separate in the first stage of meiosis. [2]

- (b) The plant was selfed. Give the phenotypic ratio of the offspring. Explain your answer.

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.....[3]

The plant is also heterozygous for another characteristic, the height of the plant. Allele T = tall is dominant to allele t = dwarf. A cross was made between the tall, red-flowered plant with a dwarf, yellow-flowered plant and produced a large number of offspring. **Table 5.1** shows the results.

Table 5.1

Phenotype	Number of offspring
Tall, red-flowered plant	35
Tall, yellow-flowered plant	14
Dwarf, red-flowered plant	16
Dwarf, yellow-flowered plant	33

- (c) The ratio of phenotypes expected in a cross such as this is 1:1:1:1. Chi-squared test was performed on these data giving a calculated value of χ^2 of 14.64.

Table 5.2: Distribution of χ^2

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (i) Use the calculated value of χ^2 and the table of probability provided in **Table 5.2** to find the probability of the results of the cross is due to chance.

Probability[1]

(ii) State what conclusions may be drawn from the probability found in (c)(i).

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.....[2]

(iii) Explain the difference between the expected and actual results of the test cross.

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.....[3]

(d) Briefly explain how another factor, besides crossing over, can bring about genetic variation in the offspring produced by a sexually reproducing organism (Exclude mutation from your answer).

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.....[1]

[Total: 12]

- 6 (a) Outline the events that will occur when an action potential reaches the pre-synaptic knob of a cholinergic neuron resulting in an action potential in the post-synaptic neurone.

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.....[3]

- (b) GABA is a neurotransmitter which inhibits the production of action potentials. **Fig. 6.1** shows how the release of GABA from a pre-synaptic membrane affects the membrane potential of a post-synaptic membrane.

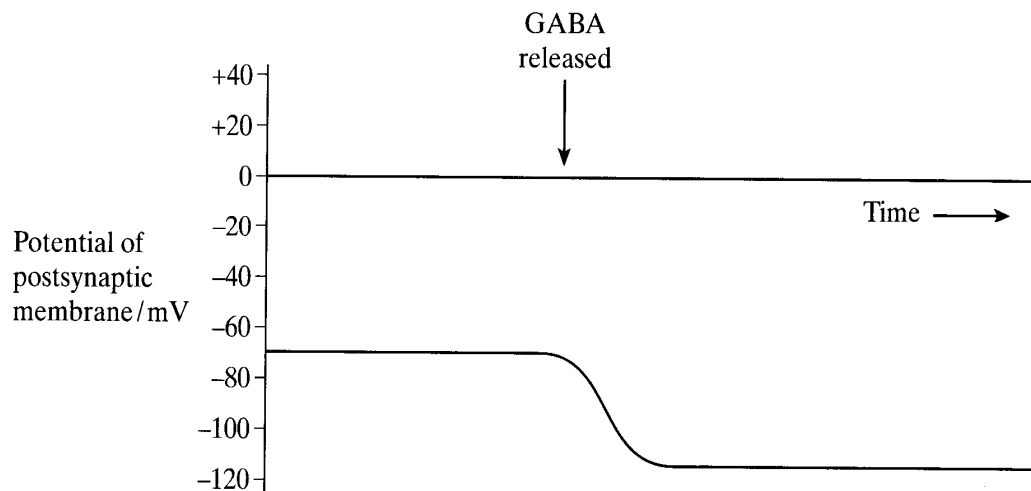


Fig. 6.1

(b) When the post-synaptic membrane is stimulated by acetylcholine, an action potential is less likely if GABA is released at the same time. Explain why.

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.....
.....[2]

(c) State two ways in which a post-synaptic potential differs from an action potential.

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.....[2]

(d) List the main differences between nervous and endocrine coordination.

1.
.....2.
.....
.....[2]

(e) In certain neuron cells, small peptide neurotransmitters are used instead of acetylcholine. These enkephalin molecules are synthesized at the cell body of the neurons.

Outline the events that will occur after enkephalin is synthesized until they are found at the pre- synaptic knob.

.....
.....
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.....[2]

[Total: 11]

- 7 HIV, like any evolving entity, has been deeply marked by its history. There are two type of HIV: HIV-1 and HIV-2. Both are sexually transmitted and are clinically indistinguishable. Worldwide the predominant virus is HIV-1. HIV-2 is rare and is concentrated in West Africa.

HIV can also be divided into 3 groups: M, N and O. Group O appears to be restricted to west-central Africa and has 4 subtypes (1-4). Group N was discovered in Cameroon. More than 90% of HIV-1 infection and belong to HIV-1 group M. Within M there are at least 10 subtypes (A-J).

Scientists studying the evolutionary history of HIV found that it is closely related to other viruses.

The diagram shows some of the evolutionary history of HIV as we know it today. An ancestral virus (bottom) evolved into strains that infected chimpanzees (SIV). Over time, new strains began to infect humans (HIV).

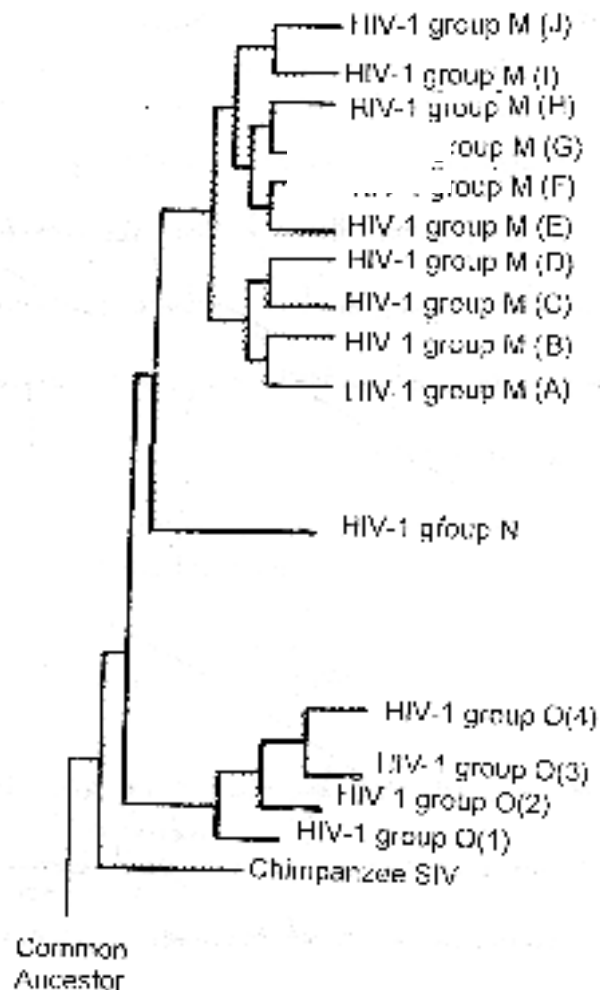


Fig 7.1

(a) Describe briefly how the phylogenetic tree shown in **Fig 7.1** is constructed.

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.....[2]

(b) Explain the significance of the branching points in the phylogeny.

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.....[2]

(c) Suggest a reason for the large number of HIV-1 groups and subtypes.

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.....[2]

(d) State the evolutionary pattern shown by SIV and HIV virus.

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.....[1]

- (d) Nucleotide substitutions in genes coding for proteins can either be silent or non-silent. In silent substitutions of a gene coding for a protein, the amino acid sequence produced remains unchanged. The table below shows the rate of non-silent (NS) to silent (S) nucleotide substitutions for 4 genes found in the HIV virus (HIV-1 strain).

Gene	NS/S
Capsid	0.822
RT	2.110
GG2A	4.574
GG4B	1.851

Table 7.2

- (i) Which of the genes would give the highest variation among the viruses?

.....[1]

- (ii) If you were a scientist developing a drug or vaccine against HIV-1, explain which of the genes would you target, with reference to **Table 7.2**?

.....

[2]

[Total: 10]

Section B

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

8 (a) Describe the differences between continuous and discontinuous variation.[6]

(b) Discuss the sources of variation important for natural selection to occur. [7]

(c) Explain how natural selection may bring about evolution. [7]

9 (a) Explain the principles of homeostasis with reference to regulation of blood glucose concentration by insulin. [10]

(b) Describe the response of liver cells in the presence of glucagons, with reference to the stages of cell signaling. [10]